

**What makes an  
asset the  
dominant  
intermediary?**



**Centre for Blockchain  
Technologies**

# **The Making of Dominant Vehicle Currencies**

Evidence from DeFi  
Jiahua Xu

# Transaction-level route structures

## Direct swap

hash  $h_D$ ; component 0



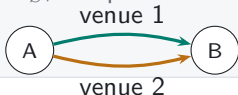
## Sequential vehicle route

hash  $h_V$ ; component 0



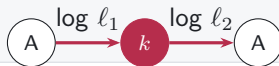
## Parallel split

hash  $h_S$ ; component 0



## Round trip

hash  $h_R$ ; component 0

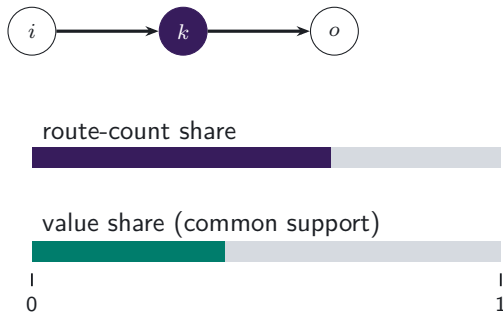


The hash groups swap events. Token connectivity assigns components, and log index orders legs within the transaction.

# Measuring vehicle dominance

## MEASUREMENT

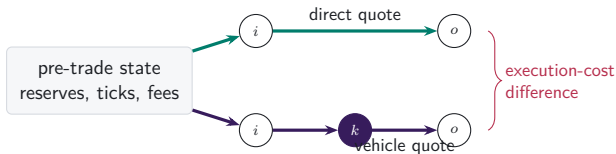
- Dominance is the share of indirect trade intermediated by asset  $k$ .
- Count share treats each route equally; value share weights routes by dollar value.
- Network measures capture how broadly  $k$  connects endpoint pairs.



# The blockchain reveals chosen routes and contemporaneous alternatives

## RESEARCH DESIGN

- FX data typically reveal executed trades.
- On-chain reserves and fees also let us price feasible alternatives.
- The comparison holds market conditions fixed at execution.

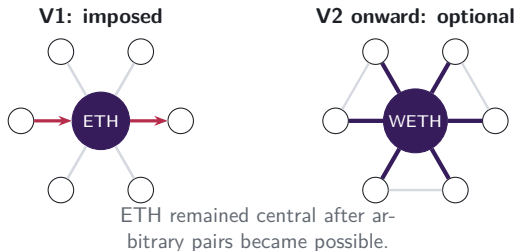


# Protocol design first imposed ETH intermediation

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## INSTITUTIONAL FACT

- Uniswap V1 created one exchange contract per token, paired with ETH.
- Token-to-token swaps therefore moved through ETH by construction.
- V2 allowed arbitrary pairs; the ETH-centred pairing network persisted.



# The panel follows the market across six years

DATA

2,332

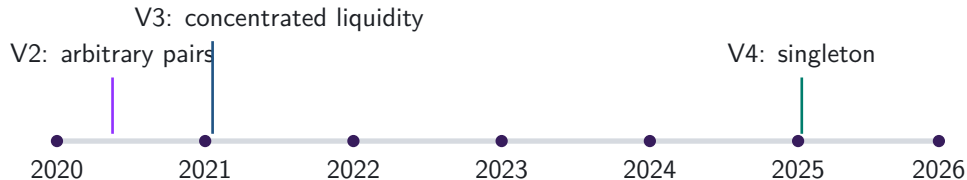
calendar days

471.6m

swap legs in reconstructed  
routes

8

DEX deployments



Routes are reconstructed transaction by transaction, then aggregated to daily and weekly observations.

# Four measures capture use, scale, and network position

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## 1. Vehicle share

Fraction of route units intermediated by asset  $k$ .

## 2. Excess use

Intermediary share relative to endpoint demand.

## 3. Network position

Degree, strength, and betweenness on the trading graph.

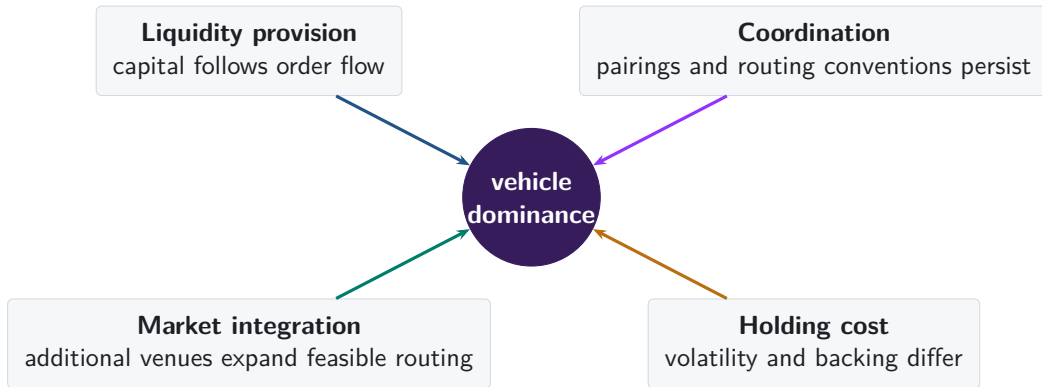
## 4. Execution advantage

Direct-route output relative to a feasible vehicle route at the same pre-trade state.

**Together, the measures distinguish use, scale, network position, and execution cost.**

# Four forces can sustain vehicle dominance

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# The empirical design separates formation, rotation, and persistence

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## FORMATION

### Architecture changes

Imposed intermediation, arbitrary pairing, concentrated liquidity, and singleton settlement.

## ROTATION

### Panel variation

Count share, value share, excess use, network position, and route complexity over time.

## PERSISTENCE

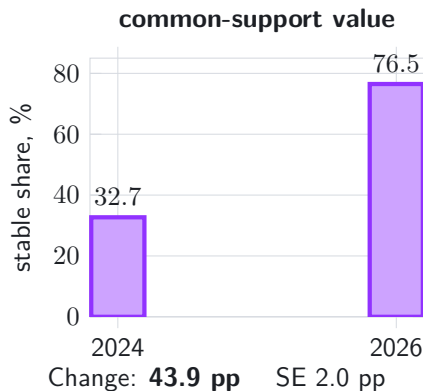
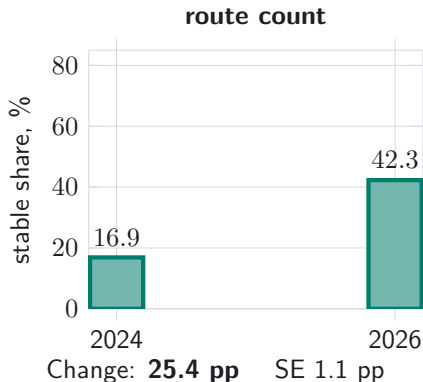
### Contemporaneous alternatives

An incumbent retains volume despite a cheaper feasible route.



# Stablecoins gained vehicle share by count and by value

## DESCRIPTIVE RESULT

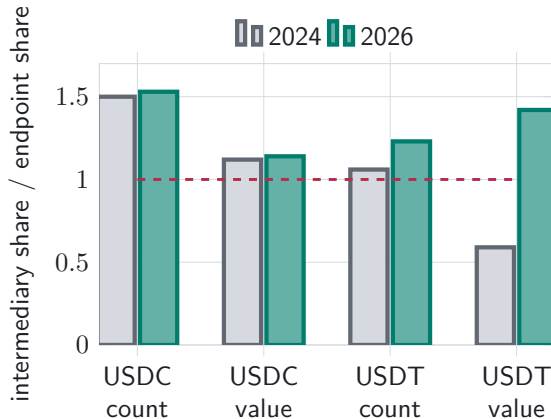


Observed two-leg routes with comparable dollar values; daily shares are equally weighted within endpoint-year.

# USDT drives the shift toward stablecoin intermediation

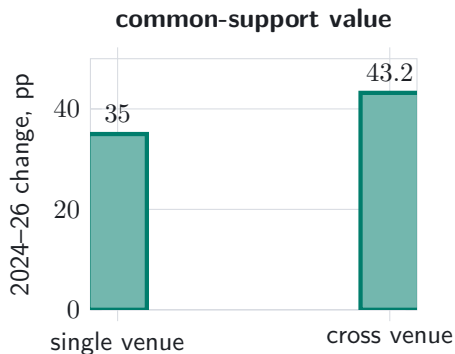
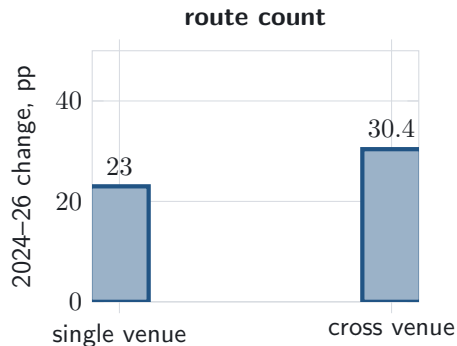
## TOKEN DECOMPOSITION

- USDC was already used more as an intermediary than its endpoint share implies.
- USDT shifted from route endpoints toward intermediation.
- USDC and USDT account for 92.1% of the stablecoin count-share increase.



# Rotation is present within venues and stronger across venues

## MARKET INTEGRATION



The increase is **7.5 pp** larger for cross-venue routes by count (SE 1.3) and **8.2 pp** larger by value (SE 2.3).

# Dominance reflects coordination, demand, and market design

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1. **A protocol can create a vehicle role.**

V1 imposed ETH intermediation; the ETH-centred pairing network persisted after arbitrary pairs became possible.

2. **The dominant vehicle can rotate.**

Stablecoins gained 25.4 percentage points by route count and 43.9 points by common-support value from 2024 to 2026.

3. **USDT drives the transition.**

Its intermediary share rose above its endpoint share, and the rotation is stronger in cross-venue routing.

# Appendix map

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## Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

## Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

## Robustness

- A7 Daily and weekly frequencies
- A8 Count and value weighting
- A9 Endpoint demand

## Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

## A1. One reconstructed route is one unit

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**one route unit**  
two swap legs

- Each linked  $i \rightarrow k \rightarrow o$  route is counted once.
- Asset  $k$  is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by  $k$ .

## A2. Backing regimes vary over time

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### **Native platform asset**

ETH, WETH; volatile settlement asset and protocol-native demand.

### **Fiat-reserve stablecoin**

USDC, USDT; reserve and redemption design.

### **On-chain collateralised**

DAI, LUSD, crvUSD; collateral, liquidation, and backing changes.

### **Tokenised external asset**

WBTC, tokenised gold; wrapped claim on an external asset.

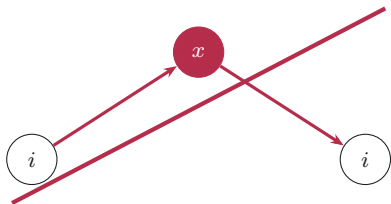
Heterogeneity tests use each asset's contemporaneous backing regime.



### A3. Sample restrictions preserve value measurement

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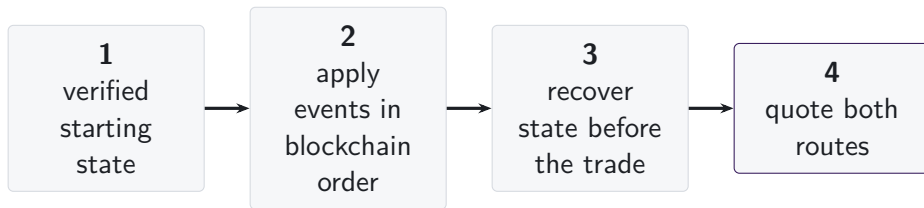
- Round trips are excluded from intermediation measures.
- Dollar values require consistent prices and verified token identities.
- We retain routes whose two legs imply dollar values within the pre-specified tolerance.



round trip: excluded from  
value-weighted vehicle measures

## A4. State is reconstructed immediately before execution

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- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

## A5. AMM designs require different state variables

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### **Constant product**

Reserves, fee, invariant. **Concentrated**

### **liquidity**

Active liquidity, price, tick, and initialised-tick changes. **StableSwap**

### **families**

Balances, amplification, rates, and oracle state.

Estimates include protocol families whose balances, fees, and pricing state can be independently validated.

### **Weighted pools**

Vault balances, weights, scaling, and rate providers. **V4 singleton**

Pool key, tick state, and fee for standard pools without custom hooks.

## A6. V1 forced routes have an on-chain signature

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- Token-to-token execution emits two linked swaps through ETH.
- The paired legs report equal ETH quantities in 87.4% of identified forced routes.
- This two-leg signature identifies protocol-imposed intermediation.



**217,003**  
forced routes

## A7. Daily and weekly frequencies answer different questions

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- |                                  |                                                     |
|----------------------------------|-----------------------------------------------------|
| 1. <b>Primary panel</b>          | day observations; date fixed effects                |
| 2. <b>Calendar sensitivity</b>   | day observations; week and weekday fixed effects    |
| 3. <b>Aggregation robustness</b> | complete weeks; week fixed effects                  |
| 4. <b>Transition series</b>      | exact 1-, 7-, 30-, and 120-day links; HAC inference |

Weekly ratios are formed from weekly sums; results are reported for weeks ending on each day of the week.

## A8. Count and value define separate dominance margins

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### Count-weighted share

$$S_{k,t}^N = \frac{N_{k,t}^I}{\sum_j N_{j,t}^I}$$

Each reconstructed route carries one unit of weight.

### Value-weighted share

$$S_{k,t}^V = \frac{\text{Vol}_{k,t}^I}{\sum_j \text{Vol}_{j,t}^I}$$

Each route carries reliably priced dollar value on common support.

Differences between count and value shares may reflect trade size, route structure, or the mix of intermediary assets.

## A9. USDT gains intermediary use beyond endpoint demand

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### Route-count share

- Intermediary-minus-endpoint share rises by **2.39 pp**.
- HAC s.e.: 0.89 pp.

### Dollar-value share (common support)

- Gap moves from  $-7.13$  **pp** to  $+8.14$  **pp**.
- Change: 15.27 pp; HAC s.e.: 1.50 pp.

**USDT's intermediary share rises above its endpoint share.**

## A10. Capital, inventory, and depth are distinct quantities

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### **Deposited capital**

Funds committed to a pool; on-chain balances or independently audited TVL.

### **Inventory**

Exact token amounts held by the pool.

### **Executable depth**

Amount that can be traded for a given price impact under current pool state.

### **Quote quality**

Execution price for trade size  $q$  along feasible routes.

We measure capital with TVL and execution depth from the pool's pricing function.



# A11. Integration changes both reach and route choice

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## Available trading opportunities

- active venues
- feasible paths
- available intermediary assets

## Route choice conditional on availability

- direct versus intermediated
- single- versus cross-venue
- sequential versus split routing

We test whether stablecoin intermediation rises more when additional venues and routes are available, conditional on endpoint pair and date.

## A12. References: vehicle currencies and market structure

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## A13. References: decentralised exchange design

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